

02_eda.R

Complete source code for reference. The original .R file must remain in GitHub for execution.

```
# Exploratory analysis and visual outputs
data <- read.csv("outputs/clean_advertising_campaigns.csv", stringsAsFactors = FALSE)
dir.create("outputs/plots", showWarnings = FALSE)
platform <- aggregate(cbind(spend, revenue, clicks, conversions, impressions) ~ platform, data, sum)
platform$ROAS <- platform$revenue / platform$spend
platform$CTR <- platform$clicks / platform$impressions
platform$CVR <- platform$conversions / platform$clicks
write.csv(platform, "outputs/platform_summary.csv", row.names = FALSE)

png("outputs/plots/roas_by_platform.png", width=1000, height=600)
barplot(platform$ROAS, names.arg=platform$platform, las=2, ylab="ROAS", main="Return on Ad Spend by Platform")
dev.off()

data$month <- format(as.Date(data$date), "%Y-%m")
monthly <- aggregate(cbind(spend, revenue, conversions) ~ month, data, sum)
monthly <- monthly[order(monthly$month),]
write.csv(monthly, "outputs/monthly_summary.csv", row.names = FALSE)
png("outputs/plots/monthly_revenue.png", width=1000, height=600)
plot(seq_len(nrow(monthly)), monthly$revenue, type="o", xaxt="n", xlab="Month", ylab="Revenue", main="Monthly Advertising Revenue")
axis(1, seq_len(nrow(monthly)), monthly$month, las=2, cex.axis=.7)
dev.off()
```